

AI FAQ Assistant API Project

Python + Flask Backend

1. Project Files

```
AI-FAQ-Assistant/  
├──  
├── app.py  
├── faq_data.json  
├── requirements.txt  
├── .env  
└── README.md
```

2. app.py

```
from flask import Flask, request, jsonify  
from openai import OpenAI  
from dotenv import load_dotenv  
import os  
import json  
  
load_dotenv()  
  
app = Flask(__name__)  
  
client = OpenAI(api_key=os.getenv("OPENAI_API_KEY"))  
  
with open("faq_data.json", "r", encoding="utf-8") as file:  
    faq_data = json.load(file)  
  
FAQ_TEXT = "\n".join(  
    f"Q: {item['question']}\nA: {item['answer']}"  
    for item in faq_data  
)  
  
@app.route("/")  
def home():  
    return "AI FAQ Assistant API is running."  
  
@app.route("/api/ask", methods=["POST"])  
def ask_question():  
    data = request.get_json()  
  
    question = data.get("question", "").strip()  
  
    if not question:  
        return jsonify({"error": "Question is required"}), 400  
  
    prompt = f"""  
You are an AI FAQ Assistant.  
  
Answer the user's question using the FAQ information below.
```

```
FAQ_DATA:
{FAQ_TEXT}
```

Rules:

1. Give a simple and clear answer.
2. If the FAQ does not contain the answer, say:
"Sorry, I don't have information about that."
3. Do not invent information.

User Question:

```
{question}
"""
```

```
        response = client.responses.create(
            model="gpt-5.6-luna",
            input=prompt
        )

        return jsonify({
            "question": question,
            "answer": response.output_text
        })
```

```
if __name__ == "__main__":
    app.run(debug=True)
```

3. `faq_data.json`

```
[
    {
        "question": "What is the college working time?",
        "answer": "The college working time is 9:00 AM to 4:30 PM."
    },
    {
        "question": "How can I contact the college office?",
        "answer": "You can contact the college office during working hours."
    },
    {
        "question": "What courses are available?",
        "answer": "The college offers undergraduate and postgraduate courses."
    },
    {
        "question": "How do I apply for admission?",
        "answer": "Students can contact the admission office for application
details."
    }
]
```

4. `requirements.txt`

```
flask
openai
python-dotenv
```

5. .env

OPENAI_API_KEY=your_api_key_here

6. Important Note

Keep the API key in an environment variable rather than putting it directly into the source code.

7. Test API

Install the required packages and run the application:

```
pip install -r requirements.txt
python app.py
```

API endpoint:

`http://127.0.0.1:5000/api/ask`

POST request:

```
{
  "question": "How can I contact the college office?"
}
```

8. Example Response

```
{
  "question": "How can I contact the college office?",
  "answer": "You can contact the college office during working hours."
}
```

9. Future Enhancement

The project can be extended with an HTML/CSS frontend, database integration, authentication, FAQ administration, document upload, and a richer AI question-answering interface.